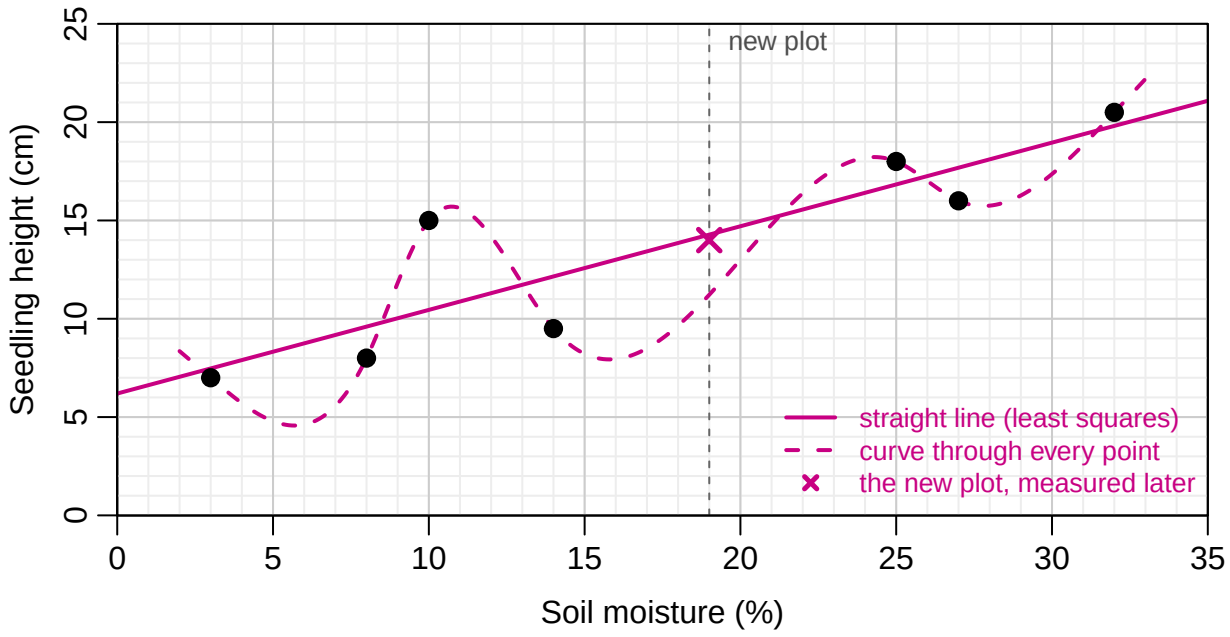
NRES 746, Week 10, Lecture 1: Connect the Dots

Answer key. Answers are in magenta sans-serif type; longer answers have a bar in the margin.

Instructions: Work alone on Part 1 (about 3 minutes), then compare with a neighbor for Part 2. Keep laptops closed.

The data

Seedling height after one growing season in seven restoration plots that differ in soil moisture.



Part 1: Two models (individual)

1a. On the figure, draw (i) a straight line that fits the seven points well, and (ii) a smooth curve that passes through every point.

> One example of each is drawn above; yours will differ somewhat.

1b. Which of your two models has the smaller sum of squared residuals for these seven plots? The curve: its residuals are all 0 (the least-squares line's sum is 35.2).

1c. A new plot has 19% soil moisture (dashed line). Read off each model's predicted height. Straight line: about 14.3 cm. Curve: about 11.2 (for the curve above) cm. Which prediction would you bet on, and why?

> The line. No plots were measured between 14% and 25%, so the curve's path there comes only from forcing it through the high plot at 10% and the low one at 14%, which is more likely noise than a real effect of moisture. The new plot measured 14 cm.

Part 2: Compare (with a neighbor)

2a. Lay your figures side by side. Which agree more closely between the two of you, your straight lines or your curves? What does that tell you about judging a model only by how well it fits the data used to fit it?

> The straight lines agree closely. The curves match at the seven points but wander differently between them. A curve through every point always fits perfectly, so fit to the same data can't tell a good model from an overfit one; we need a penalty for complexity or new data to test against.